INCH-POUND

MS51815H
30 March 2022
SUPERSEDING
MS51815G
w/AMENDMENT 1
10 October 2018

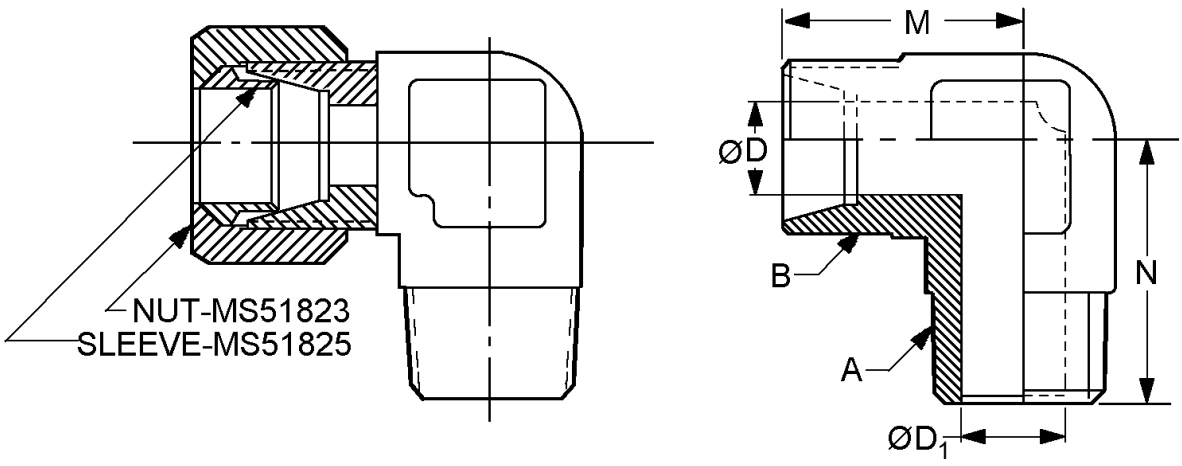
DETAIL SPECIFICATION SHEET

ELBOW PIPE TO TUBE 90 DEGREE, MALE PIPE END, FLARELESS TYPE, HYDRAULIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated after 3 April 2018 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-18866.



See notes at end of figure.

FIGURE 1. Elbow 90°, pipe to tube.

AMSC N/A
FSC 4730

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Standard sizes	Dash number	Tube OD Nom inches (mm)	A Thread NPTF	B (See note 8)
Assembly	Body (see note 5)			
1	31	.1250 (3.175)	.125 -27	.3125 - 24UNF
2	32	.1875 (4.763)	.125 -27	.3750 - 24UNF
3	33	.2500 (6.350)	.125 -27	.4375 - 20UNF
4	34	.3125 (7.938)	.125 -27	.5000 - 20UNF
6	36	.3750 (9.525)	.250 - 18	.5625 - 18UNF
8	38	.5000 (12.700)	.375 - 18	.7500 - 16UNF
12	42	.6250 (15.875)	.500 - 14	.8750 - 14UNF
14	44	.7500 (19.050)	.750 - 14	1.0625 - 12UN
15	45	.8750 (22.225)	.750 - 14	1.1875 - 12UN
18	48	1.0000 (25.400)	1.000 - 11 1/2	1.3125 - 12UN
20	50	1.2500 (31.750)	1.250 - 11 1/2	1.6250 - 12UN
21	51	1.5000 (38.100)	1.500 - 11 1/2	1.8750 - 12UN
22	52	2.0000 (50.800)	2.000 - 11 1/2	2.5000 - 12UN

Standard sizes	Dash number	D dia. Drill inches (mm)	D_1 Drill dia. inches (mm)	M Inches (mm) ±.03 (0.76)	N Inches (mm) ±.03 (0.76)
Assembly	Body (see note 5)				
1	31	.093 (2.36)	.188 (4.78)	.77 (19.56)	.72 (18.29)
2	32	.125 (3.18)	.188 (4.78)	.83 (21.08)	.72 (18.29)
3	33	.203 (5.16)	.188 (4.78)	.89 (22.61)	.78 (19.81)
4	34	.234 (5.94)	.188 (4.78)	.95 (24.13)	.81 (20.57)
6	36	.281 (7.14)	.281 (7.14)	1.05 (26.67)	1.09 (27.69)
8	38	.422 (10.72)	.406 (10.31)	1.25 (31.75)	1.22 (30.99)
12	42	.500 (12.70)	.531 (13.49)	1.42 (36.07)	1.47 (37.34)
14	44	.656 (16.66)	.719 (18.26)	1.58 (40.13)	1.59 (40.39)
15	45	.718 (18.24)	.719 (18.26)	1.66 (42.16)	1.69 (42.93)
18	48	.875 (22.23)	.938 (23.83)	1.73 (43.94)	1.97 (50.04)
20	50	1.093 (27.76)	1.250 (31.75)	1.89 (48.01)	2.38 (60.45)
21	51	1.344 (34.14)	1.500 (38.10)	2.02 (51.31)	2.64 (67.06)
22	52	1.813 (46.05)	1.938 (49.23)	2.45 (62.23)	3.00 (76.20)

See notes at end of figure.

FIGURE 1. Elbow 90°, pipe to tube - Continued.

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Non-standard sizes	Dash number (see notes 6)	Tube OD Nom inches (mm)	A Thread NPTF	B (See Note 8)
Assembly	Body (See note 5)			
5	35	.2500 (6.350)	.250 -18	.438 - 20UNF
7	37	.3750 (9.525)	.375 - 18	.563 - 18UNF
9	39	.6250 (15.875)	.375 - 18	.875 - 14UNF
10	40	.3750 (9.525)	.500 - 14	.563 - 18UNF
11	41	.5000 (12.700)	.500 - 14	.750- 16UNF
13	43	.7500 (19.050)	.500 - 14	1.063 - 12UNF
16	46	1.000 (25.400)	.750 - 14	1.313 - 12UNF
17	47	.8750 (22.225)	1.000 - 11 1/2	1.188 - 12UN
19	49	1.250 (31.750)	1.000 - 11 1/2	1.625 - 12UN

Non-standard sizes	Dash number (see note 6)	D dia. Drill inches (mm)	D_1 Drill dia. inches (mm)	M Inches (mm) ±.03 (0.76)	N Inches (mm) ±.03 (0.76)
Assembly	Body (See note 5)				
5	35	.203 (5.16)	.281 (7.14)	1.03 (26.16)	1.09 (27.69)
7	37	.281 (7.14)	.406 (10.31)	1.14 (28.96)	1.22 (30.99)
9	39	.500 (12.70)	.406 (10.31)	1.42 (36.07)	1.28 (32.51)
10	40	.281 (7.14)	.531 (13.49)	1.24 (31.50)	1.47 (37.34)
11	41	.422 (10.72)	.531 (13.49)	1.35 (34.29)	1.47 (37.34)
13	43	.656 (16.66)	.531 (13.49)	1.58 (40.13)	1.59 (40.39)
16	46	.875 (22.23)	.719 (18.26)	1.73 (43.94)	1.78 (45.21)
17	47	.718 (18.24)	.938 (23.83)	1.73 (43.94)	1.97 (50.04)
19	49	1.093 (27.76)	.938 (23.83)	1.89 (48.01)	2.35 (59.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-J514.
5. Assemblies are only furnished to this specification (body, nut, and sleeve). Bodies are not to be stocked, stored, or issued.
6. Non-standard sizes are not recommended for new design.
7. The drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.
8. ASME B1.1 - Unified Inch Screw Threads (UN and UNR Thread Forms)

FIGURE 1. Elbow 90°, pipe to tube - Continued.

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REQUIREMENTS:

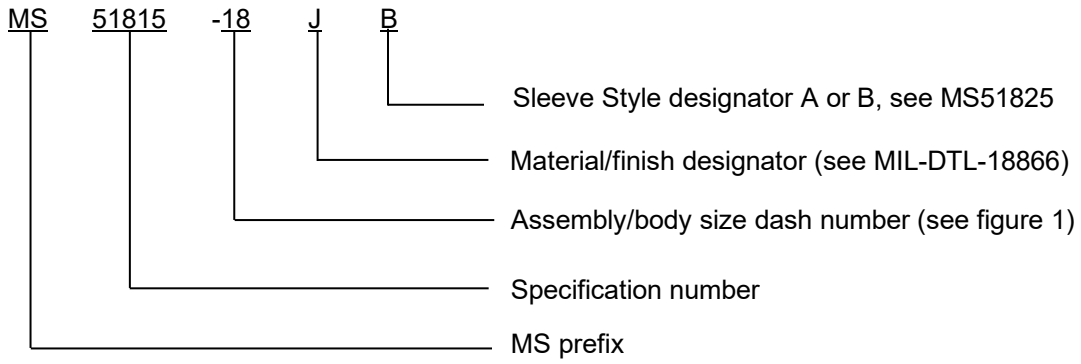
Elbow 90°, pipe to tube shall be as specified on figure 1.

Materials and finish designators shall be in accordance with MIL-DTL-18866.

Unless otherwise specified in MIL-DTL-18866, all platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Maximum operating pressure. Maximum operating pressure shall be in accordance with SAE-J514.

PIN: The PIN consists of the letters “MS”, the specification number, a dash, number for elbow pipe to tube 90° assembly/body size, and a letter for material finish designator.



PIN example: MS51815-18J indicates an elbow 90°, pipe to tube assembly, 1 inch (2.54 mm) tube OD, with 1 inch NPTF pipe threads and 1.3125 - 12UN threads, steel zinc nickel.

Marking shall be in accordance with MIL-DTL-18866.

Class I and II ozone depleting substances (ODS) shall not be used in MS51815 or any referenced procedures.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

NOTE: The following restrictions do not apply to manufacturers of the fittings and is applicable only to OEM’s and users.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

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Army and Marine ground vehicle applications using zinc plating. PIN codes FC, FD, Z, and ZC finishes have a chromate conversion coating which includes hexavalent chromium and are not recommended for Army and Marine ground systems. The use of coatings depositing hexavalent chromium for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement. For new designs PIN codes FC, FD, Z, and ZC are not to be supplied to Army and Marine ground forces. PIN code KA (MIL- PRF- 32647 Type K20) is intended as a hazardous free plating material and is a direct replacement for zinc plating with a hexavalent chromium conversion coating. PIN code KA has equivalent torque/tension behavior as the zinc plating it replaces.

Army and Marine ground vehicles. The use of cadmium plating for Army and Marine ground vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.

Army and Marine ground vehicle applications using cadmium plating. Pin code J or KB may be substituted for cadmium ("Blank" PIN code) for Army ground vehicle applications. PIN code KB (MIL-PRF-32647 Type K16) is intended as a hazardous free plating material and is a direct replacement for cadmium plating ("Blank" PIN code). PIN code KB has an equivalent torque/tension behavior as the cadmium plating it replaces.

NAVSEA shipboard owned equipment. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Table II provides a detail reference data listing for MS39202 to MS51815. This table contains coordination of equivalent tube sizes, in both compression and hydraulic type elbows. Assembly replacement is authorized.

TABLE II. Assembly equivalency (for information only).

Compression MS39202	Hydraulic MS51815	Compression MS39202	Hydraulic MS51815	Compression MS39202	Hydraulic MS51815
-1	-1	-8	-8	-15	-15
-2	-2	-9	-9	-16	-16
-3	-3	-10	-10	-17	-17
-4	-4	-11	-11	-18	-18
-5	-5	-12	-12	-19	-19
-6	-6	-13	-13	-20	-20
-7	-7	-14	-14	-21	-21

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-18866, this document references the following:

MS39202
MS51823
MS51825
ASTM B117
ASME B1.1
SAE-J514
T9070-AL-DPC-020/077-2

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force – 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2022-007)

Review activities:
Army - AR, MI
Navy - AS, CG, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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